08/10/26.

Lecture - 01.

Agenda / Syllabus

Syllabus : ① RDB ② NoSql : Mongo

- DBMS vs RDBMS
- MySQL: Introduction, Installation, …
- SQL
  - CREATE TABLE, MySQL data types
  - SELECT with LIMIT, ORDER, WHERE, GROUP BY, HAVING
  - INSERT, UPDATE, DELETE
  - Joins, Sub-queries
  - Transaction & Locking
  - GRANT & REVOKE
- MySQL programming (PSM)
  - Stored procedure
  - Cursors
  - Functions
  - Triggers

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DBMS

- Any enterprise application need to manage data.
- In early days of software development, programmers store data into files and does operation on it. However data is highly application specific.
- Even today many software manage their data in custom formats e.g. Tally, Address book, etc.
- As data management became more common, DBMS systems were developed to handle the data. This enabled developers to focus on the business logic e.g. FoxPro, DBase, Excel, etc.
- At least CRUD (Create, Retrieve, Update and Delete) operations are supported by all databases.
- Traditional databases are file based, less secure, single-user, non-distributed, manage less amount of data (MB), complicated relation management, file-locking and need number of lines of code to use in applications.

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DBMS :- Data Base management System.

* Big Organisations have Big problems :- Softwares are made for such purpose.
  - e.g. Tally, Address, book, etc.

* CRUD (Create, Retrieve, Update and Delete.)

* Databases are less secure. ⇒ like Excel we cant.
  - But we can protect while saving.
  - But someone can delete it!

→ so hence very limited security.

→ file-locking.

Less amount of data (mB).

Traditional databases.

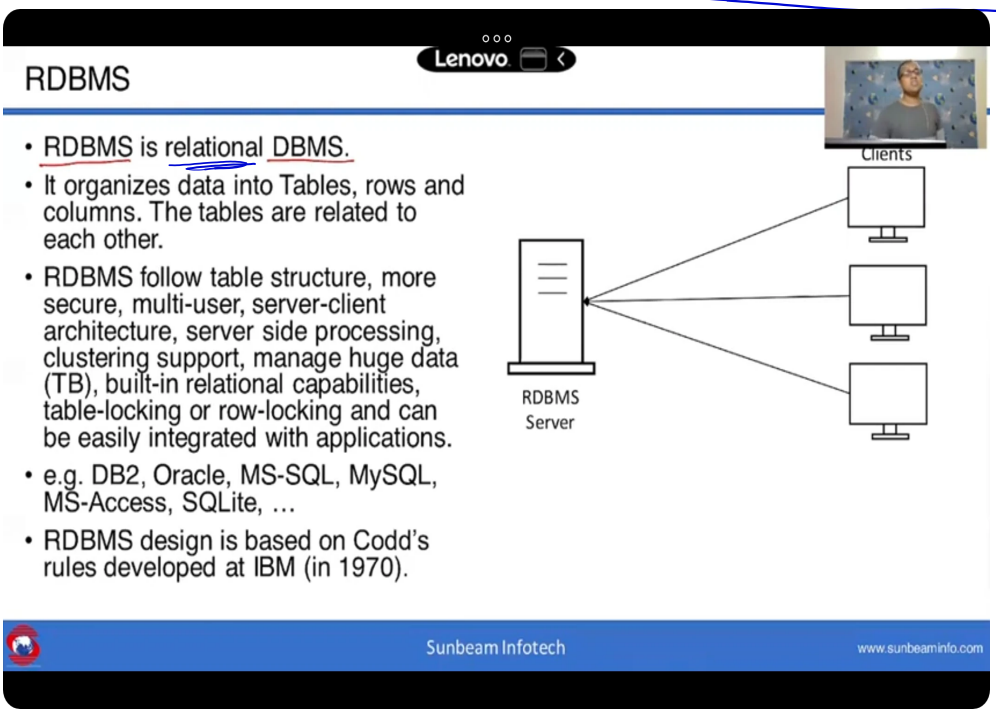
→ non distributed.

File based

less secure.

single user.

Complicated relation management.



RDBMS is also having CRUD operations but it is relational

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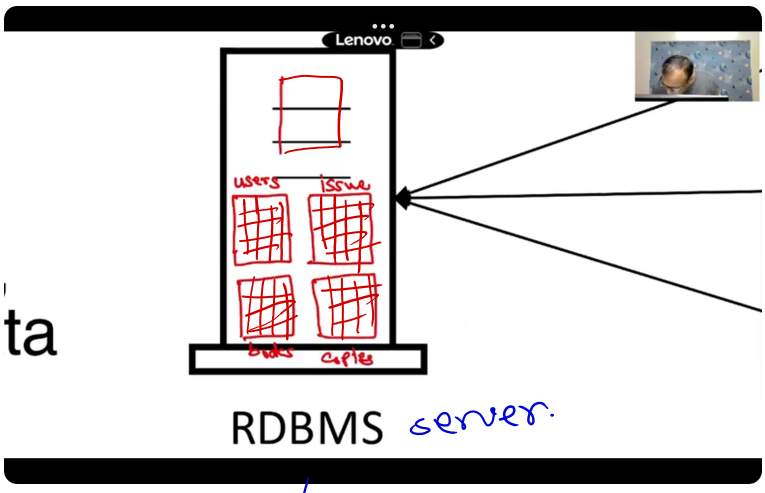
Data will be arranged in multiple tables and rows.



⇒ Instead of making different files, we will make different tables!

one user can have plenty Records.

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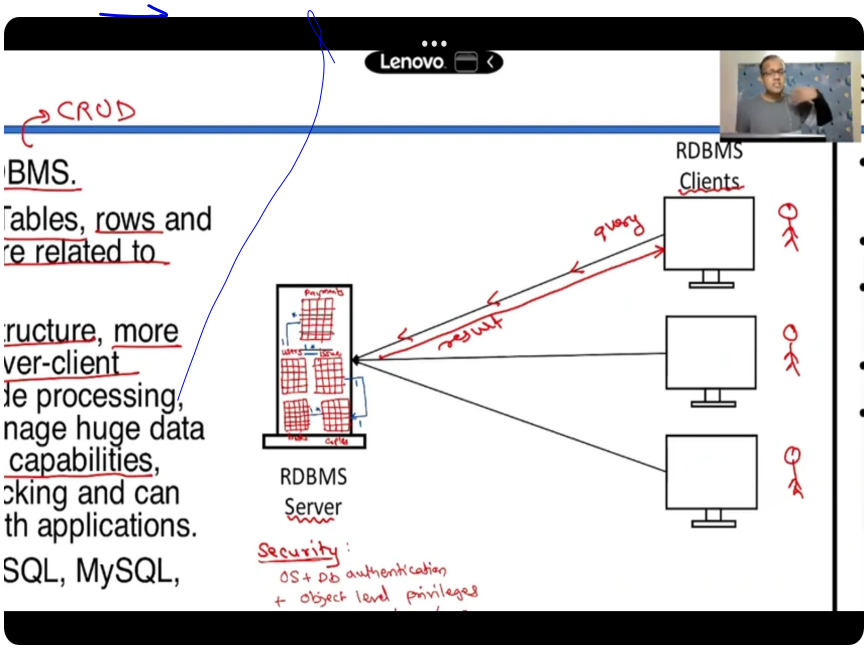
- table structure
- much more ~~se~~secure.

example:-
(OS level + Database level also)
↓
+ object level Priviledges.

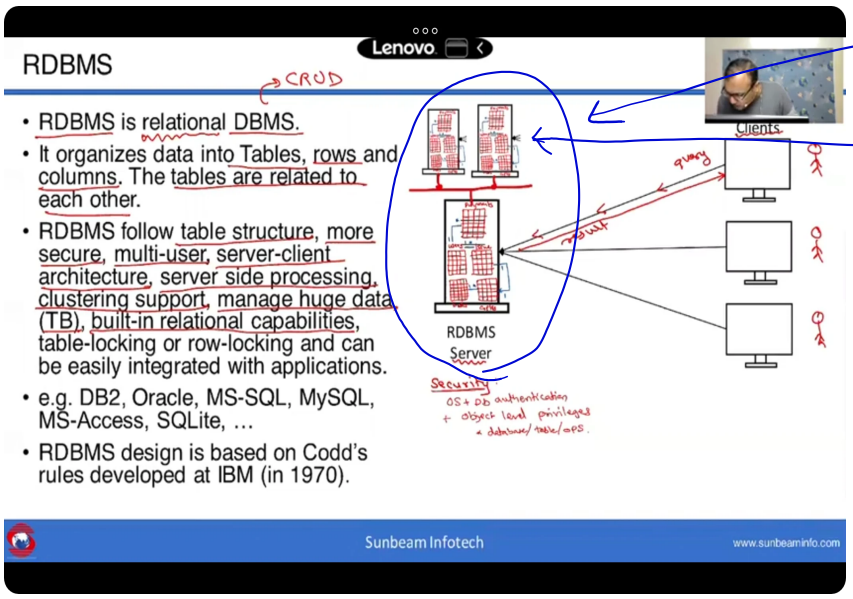
* multiple users:-

→ All data on centralised server.



- Server data process karraha hai. so advance specification is necessary.

but we are providing copy to client, so client specification no need to be high.



Database cluster

for safety multiple machines are there for backup.

* Server down ⇒ It may happen that network of all cluster is down.

* Codd's ⇒ Mathematician ⇒ 1+12 rules → based on Set theory.
  ↳ IBM.

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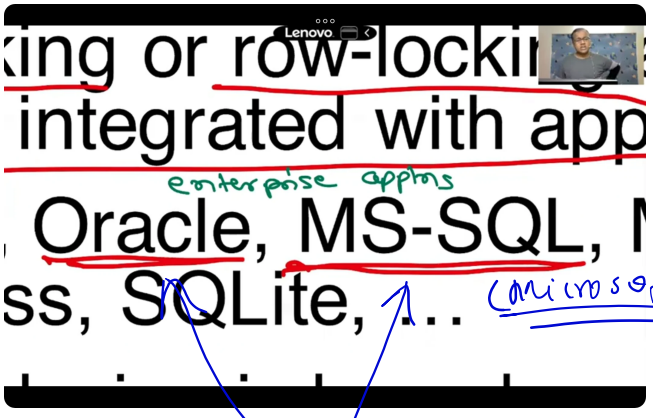
Oracle ⇒ First Database.



Both more Popular.

Both are Hearily Paid databases.

* Cobol uses DB2.

* mySQL is a partially open-source database.

* Cybase, Postgres-SQL., Informix... etc.
There are so many like these.

*. SQLite => file based RDBMS.

* All Databases have similar syntaxes. → Structured Query Language studied by all.

SQL

- Clients send SQL queries to RDBMS server and operations are perform accordingly.
- Originally it was named as RQBE (Relational Query By Example).
- SQL is ANSI standardised in 1987 and then revised multiple times adding new features. Recent revision in 2016.
- SQL is case insensitive.
- There are five major categories:
  - DDL: Data Definition Language e.g. CREATE, ALTER, DROP, RENAME.
  - DML: Data Manipulation Language e.g. INSERT, UPDATE, DELETE.
  - DQL: Data Query Language e.g. SELECT.
  - DCL: Data Control Language e.g. CREATE USER, GRANT, REVOKE.
  - TCL: Transaction Control Language e.g. SAVEPOINT, COMMIT, ROLLBACK.
- Table & column names allows alphabets, digits & few special symbols.
- If name contains special symbols then it should be back-quotes.
- e.g. Tbl1, `T1#`, `T2$` etc. Names can be max 30 chars long.

backquotes.

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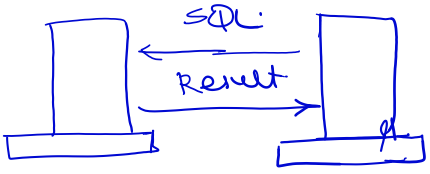

SQL :- Structured Query Language.

Case insensitive → Except table & db names on Linux/UNIX platform.

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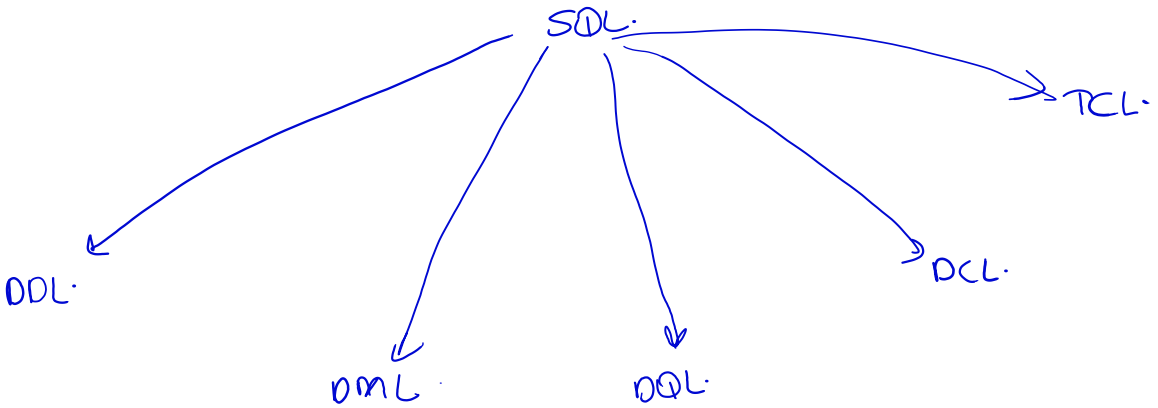

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MySQL

- Developed by Michael Widenius in 1995. It is named after his daughter [illegible].
- Sun Microsystems acquired MySQL in 2008.
- Oracle acquired Sun Microsystem in 2010.
- MySQL is free and open-source database under GPL. However some enterprise modules are close sourced and available only under commercial version of MySQL.
- MariaDB is completely open-source clone of MySQL.
- MySQL support multiple database storage and processing engines.
- MySQL versions:
  - < 5.5: MyISAM storage engine
  - 5.5: InnoDb storage engine
  - 5.6: SQL Query optimizer improved, memcached style NoSQL
  - 5.7: Windowing functions, JSON data type added for flexible schema
  - 8.0: CTE, NoSQL document store.
- MySQL is database of year 2019 (in database engine ranking).

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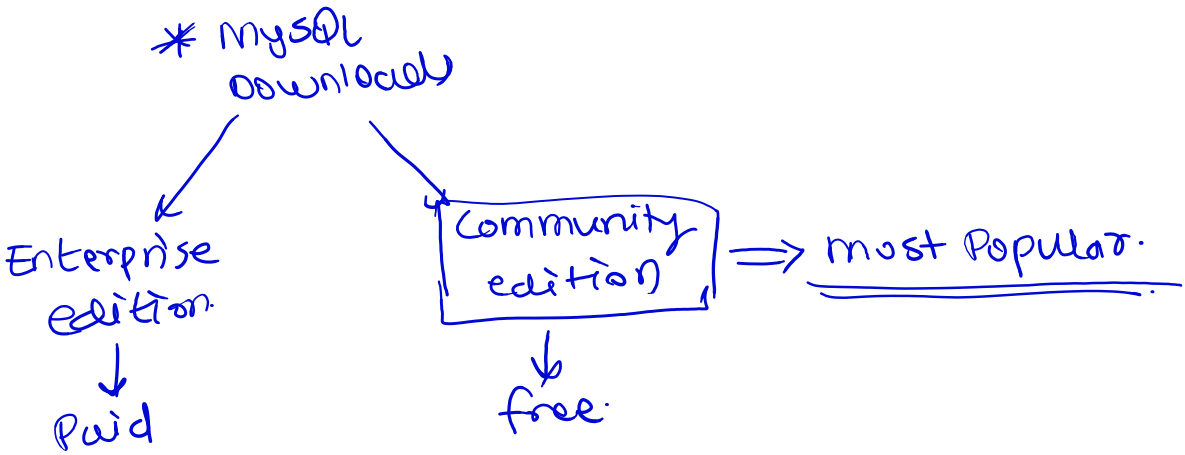
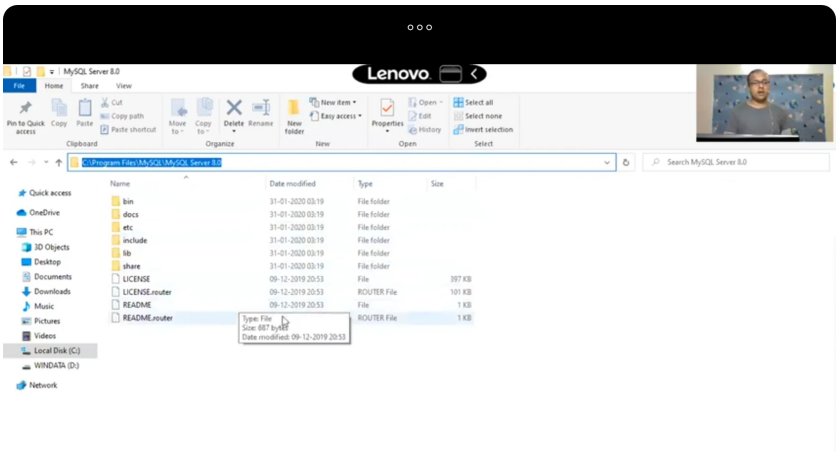
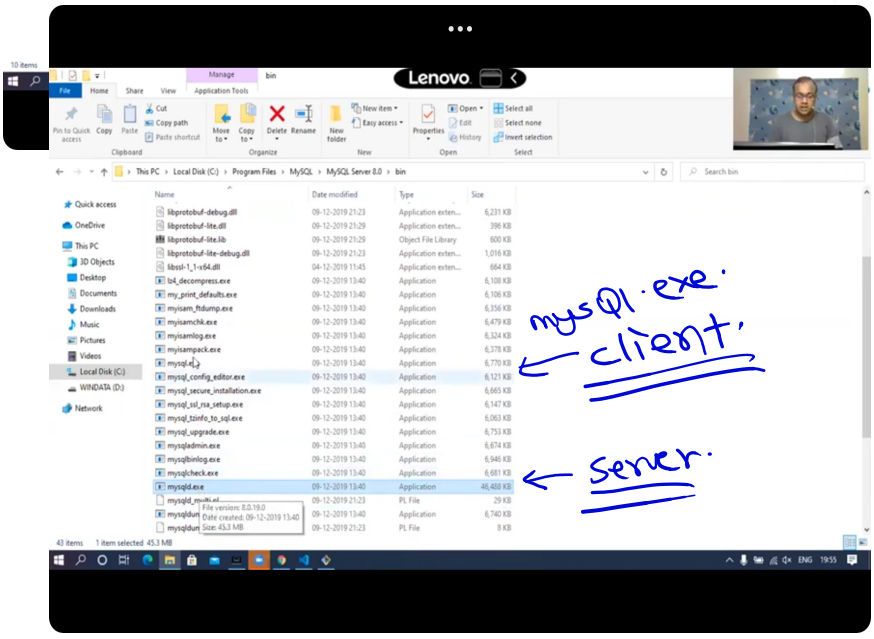
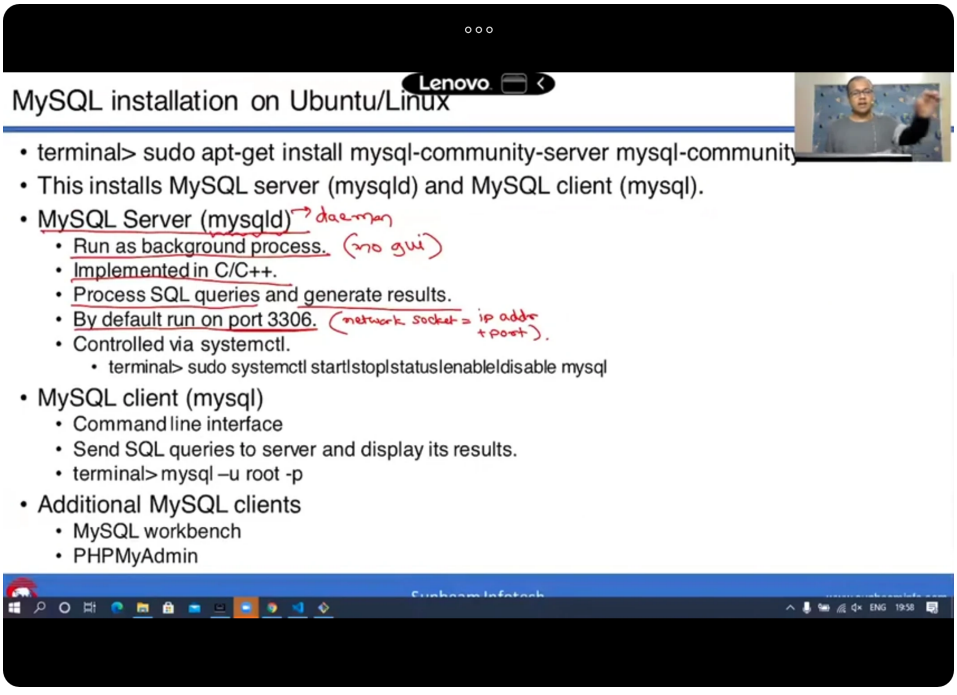


* MariaDB is open-source clone of MySQL.





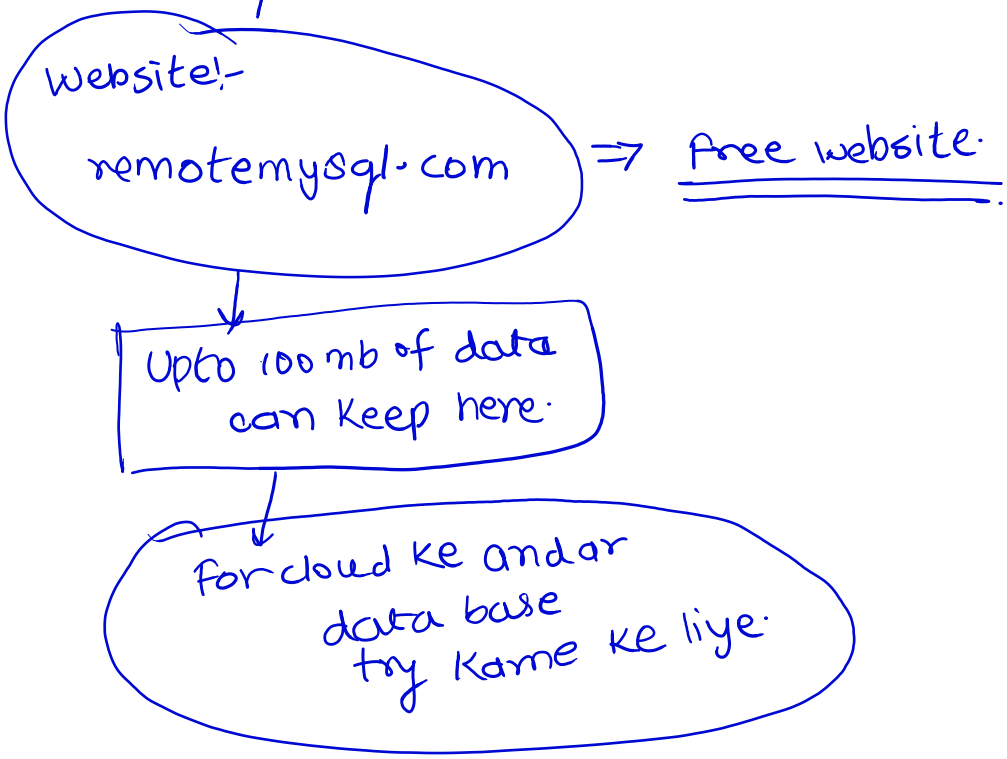
mysqld → daemon.



MySQL → Platform.
SQL → Language.

mysqld is running on server.

mysql is client.



* mySQL workbench ⟶ Desktop client.

* PHPMyAdmin ⟶ web client.

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*

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Getting started

- root login can be used to perform CRUD as well as admin operations.
- It is recommended to create users for performing non-admin tasks.
  - mysql> CREATE DATABASE db;
  - mysql> SHOW DATABASES;
  - mysql> CREATE USER dbuser@localhost IDENTIFIED BY 'dbpass';
  - mysql> SELECT user, host FROM mysql.user;
  - mysql> GRANT ALL PRIVILEGES ON db.* TO dbuser@localhost;
  - mysql> FLUSH PRIVILEGES;
  - mysql> EXIT;
- terminal> mysql –u dbuser –pdbpass
  - mysql> SHOW DATABASES;
  - mysql> SELECT USER(), DATABASE();
  - mysql> USE db;
  - mysql> SHOW TABLES;
  - mysql> CREATE TABLE student(id INT, name VARCHAR(20), marks DOUBLE);
  - mysql> INSERT INTO student VALUES(1, 'Abc', 89.5);
  - mysql> SELECT * FROM student;

Can try these all!

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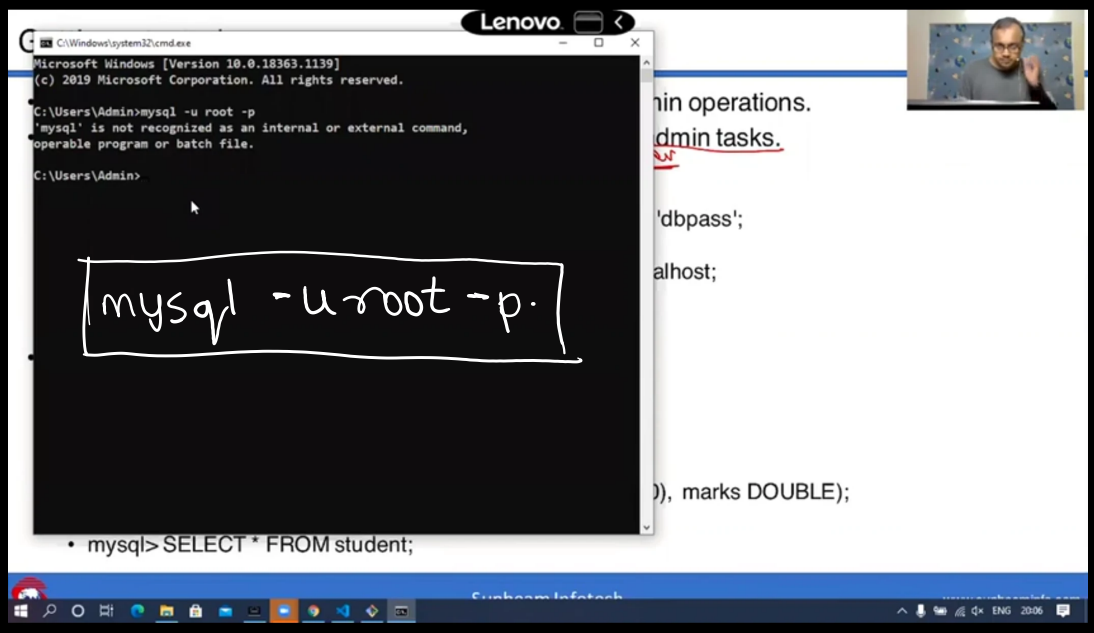


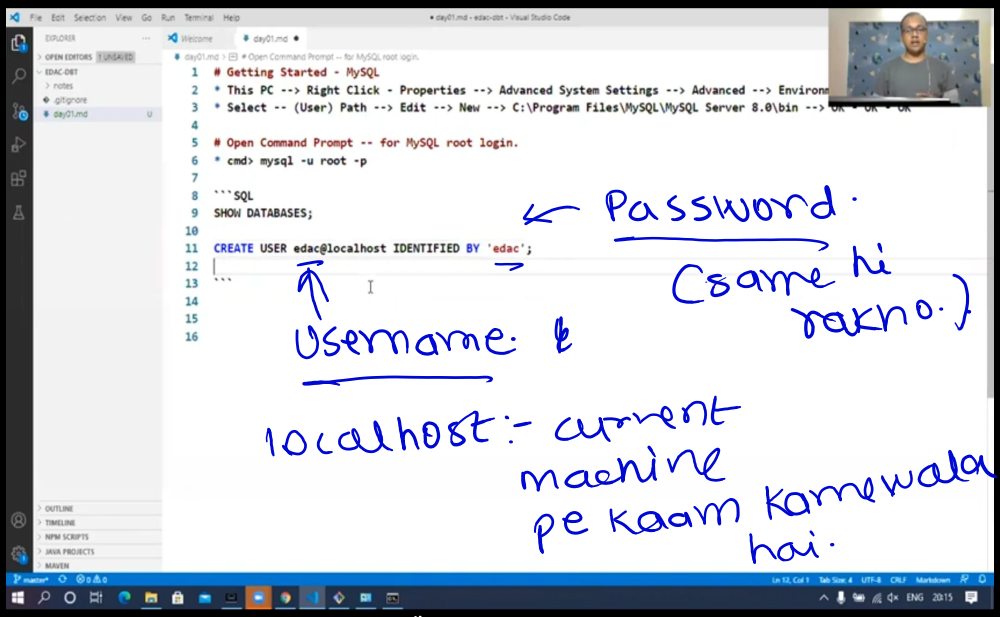
Getting Started - MySQL.

This PC → Right Click - Properties → Advanced System Settings.

Environment Variables. ← Advanced ←

Path → Edit → New → Paste the path of bin.

...



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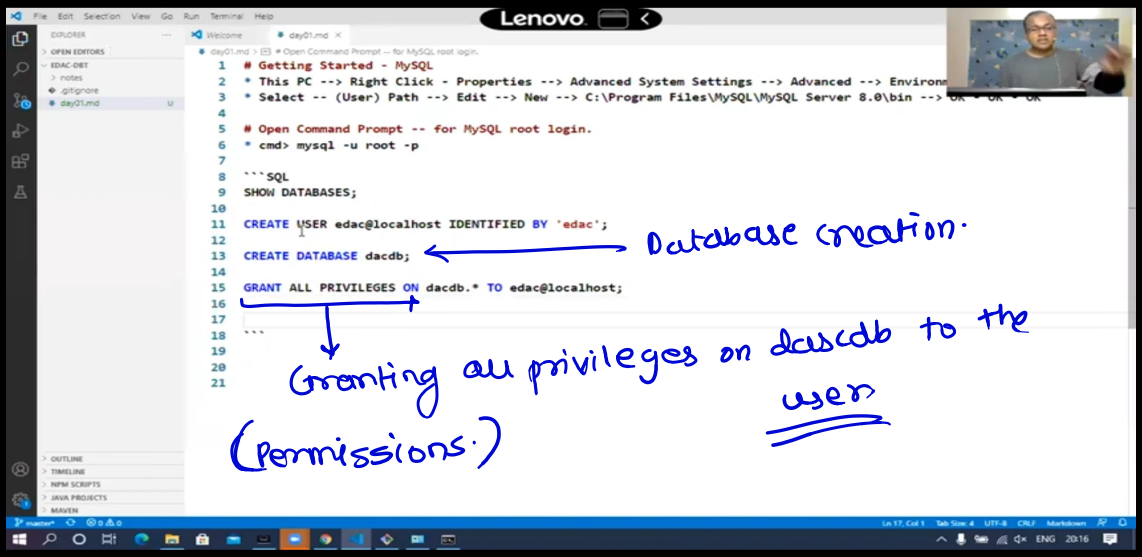


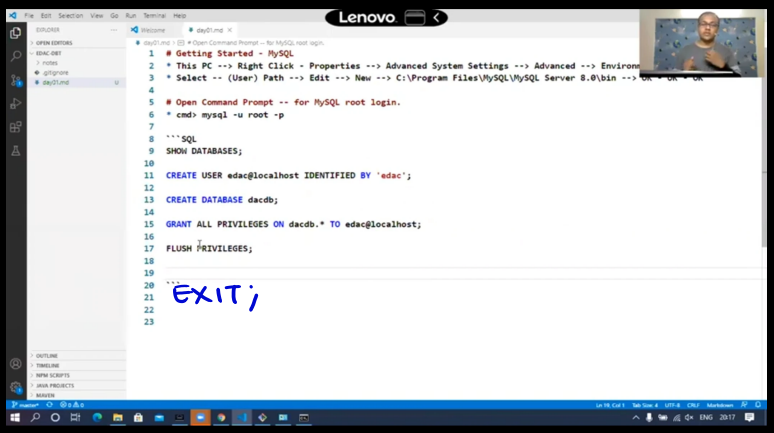
* User ab kuch bhi kar sakta hai. (Power of iternyt de di.)

* But need to activate that Power.

∴ FLUSH PRIVILEGES;

EXIT;

...



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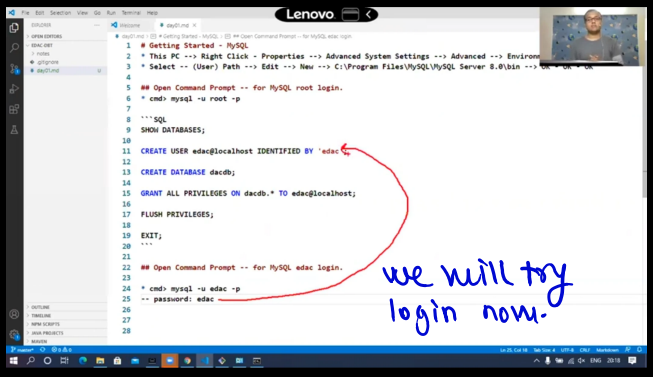

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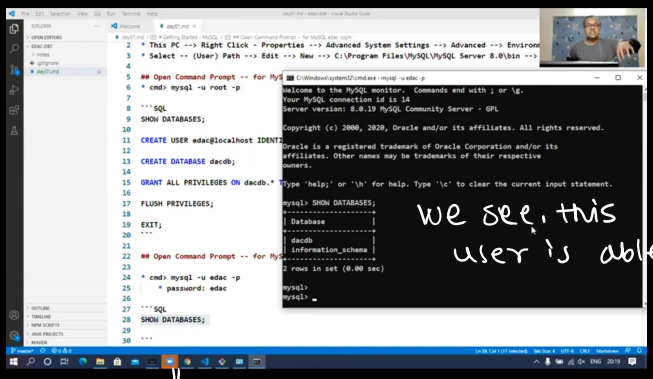

This is all basic